

Software Requirements Specification (SRS)

TastyBytes Recipe System

PHASE 1: PLANNING AND REQUIREMENT ANALYSIS

1. Introduction

1.1 Purpose

This document provides a detailed description of the requirements for the TastyBytes Recipe Management System. It is intended for academic evaluation, developers, designers, and system stakeholders.

1.2 Scope

TastyBytes is a web-based recipe management platform that allows users to create, manage, browse, and favorite recipes, with administrative moderation to ensure quality and originality.

2. Stakeholders

- Developers
- UI/UX Designers
- Testers
- General Users (Recipe Creators & Viewers)
- System Administrator
- Academic Supervisors

3. Feasibility Study

3.1 Technical Feasibility

The system is developed using frontend-focused web technologies.

Technologies Required:

- HTML5
- CSS
- Bootstrap
- PHP
- Browser localStorage (simulated backend)

3.2 Economic Feasibility

The system incurs minimal cost since all technologies are open-source. Suitable for academic and prototype deployment.

3.3 Operational Feasibility

Simple and intuitive UI allows users to easily manage recipes with minimal training.

3.4 Legal Feasibility

No sensitive data is processed beyond basic user information. The system complies with general data protection principles.

3.5 Schedule Feasibility

The project can be completed within an academic semester.

4. System Requirements

This section describes the functional and non-functional requirements of the **TastyBytes Recipe Management System**. These requirements define how the system behaves, performs, and operates to ensure reliable recipe creation, browsing, and administration.

4.1 Functional Requirements

4.1.1 User Authentication

- JavaScript (ES6+)The system shall allow users to register for an account using valid credentials.
- The system shall allow registered users to log in and log out securely.
- The system shall authenticate users before allowing access to personalized features
- The system shall assign roles to users (general user or administrator) to control access permissions.

4.1.2 Recipe Management

- The system shall allow users to create new recipes.
 - The system shall allow users to edit and delete only their own recipes.
 - The system shall require users to provide recipe details, including title, category, ingredients, and preparation steps.
 - The system shall allow users to optionally upload or attach a recipe image.
 - The system shall automatically generate a timestamp when a recipe is created or updated.
 - The system shall store newly submitted recipes with a pending approval status.
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4.1.3 Recipe Browsing and Viewing

- The system shall allow users to browse approved recipes.
- The system shall display recipes in a structured and user-friendly format.
- The system shall allow users to view detailed recipe information.
- The system shall support filtering or categorization of recipes.

4.1.4 Recipe Approval and Administration

- The system shall allow administrators to view all submitted recipes.
- The system shall allow administrators to approve or reject submitted recipes.
- The system shall update the recipe status based on the administrator's decision.
- The system shall prevent unapproved recipes from being visible to the public.
- The system shall allow administrators to remove inappropriate or duplicate recipes.

4.1.5 Favorites Management

- The system shall allow users to mark recipes as favorites.
- The system shall allow users to view their list of favorite recipes.
- The system shall allow users to remove recipes from their favorites.
- The system shall associate favorite recipes with the corresponding user account.

4.1.6 User Profile Management

- The system shall allow users to view and update their profile information
- The system shall maintain a history of recipes created by the user.
- The system shall maintain a list of recipes favorited by the user.

4.2 Non-Functional Requirements

4.2.1 Performance Requirements

- The system shall load pages within **300 milliseconds** under normal conditions.
- The system shall support multiple concurrent users without significant performance degradation.
- The system shall provide real-time updates for recipe approval status.
- The system shall ensure smooth interaction during browsing and recipe management.

4.2.2 Security Requirements

- The system shall encrypt user passwords before storage.
- The system shall use secure authentication mechanisms.
- The system shall restrict administrative actions to authorized administrators only.
- The system shall protect user data from unauthorized access.

- The system shall automatically log out inactive users after a defined period.

4.2.3 Usability Requirements

- The system shall provide a simple and intuitive user interface.
- The system shall allow users to create or submit a recipe in less than **1 minute**.
- The system shall be easy to use without requiring technical training.
- The system shall be responsive and usable across different screen sizes.

4.2.4 Reliability Requirements

- The system shall maintain an uptime of at least **99.5%**.
- The system shall ensure no loss of recipe or user data.
- The system shall perform regular data backups.
- The system shall recover gracefully from unexpected errors.

4.2.5 Scalability Requirement

- The system shall support an increasing number of users without redesign.
- The system shall allow future expansion of features.
- The system shall store large volumes of recipes and user data efficiently.

4.2.6 Compatibility Requirements

- The system shall operate correctly on modern web browsers, including Chrome, Firefox, Safari, and Edge.
- The system shall be responsive and usable on desktop, tablet, and mobile devices.

5. Use Cases

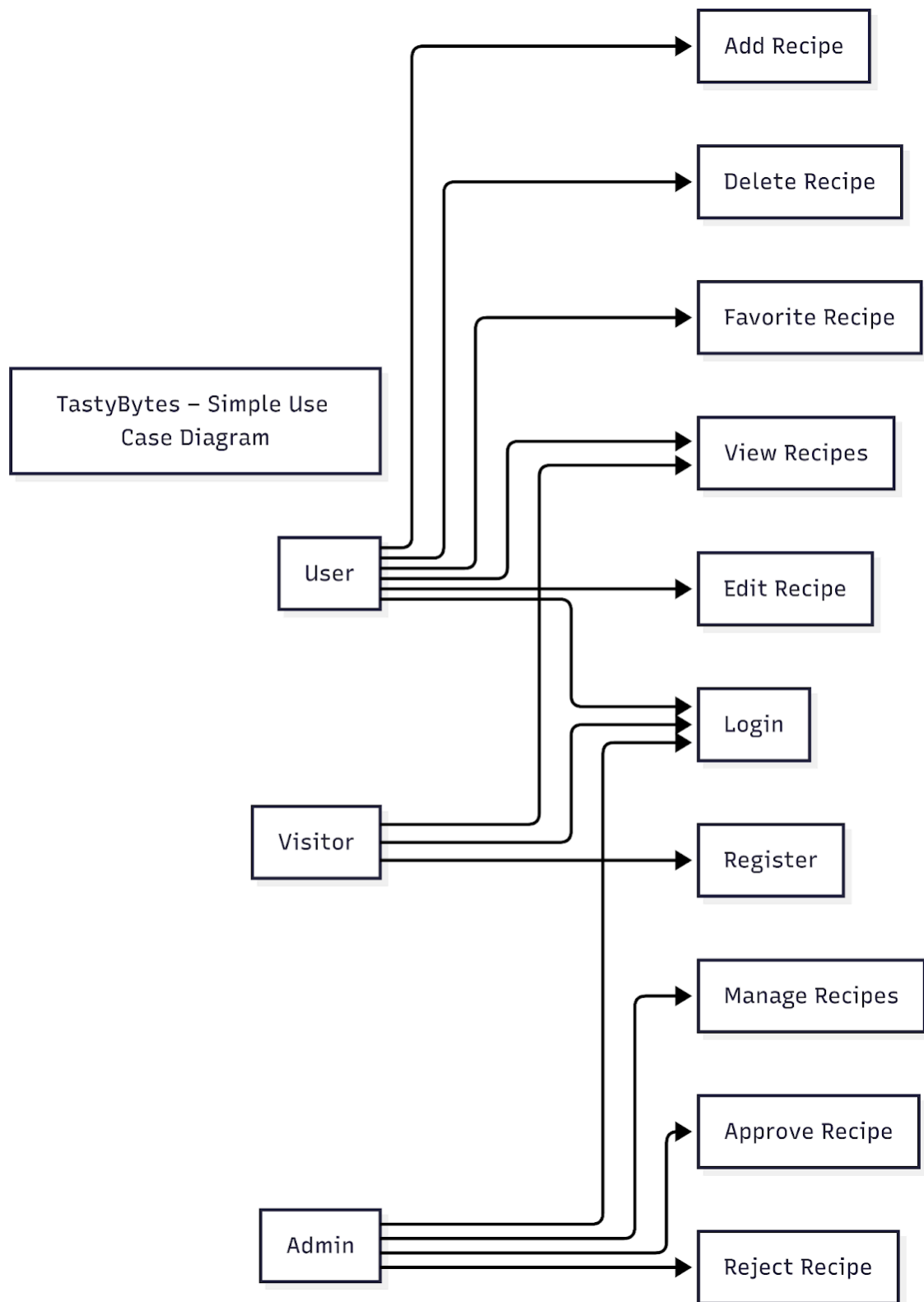
5.1 General User Use Cases

- Register
- Login/Logout
- Create Recipe
- Edit/Delete Own Recipe
- Favorite Recipe
- View Approved Recipes

5.2 Administrator Use Cases

- Login
- Approve or Reject Recipes
- Manage Users
- Remove Duplicate or Inappropriate Recipes

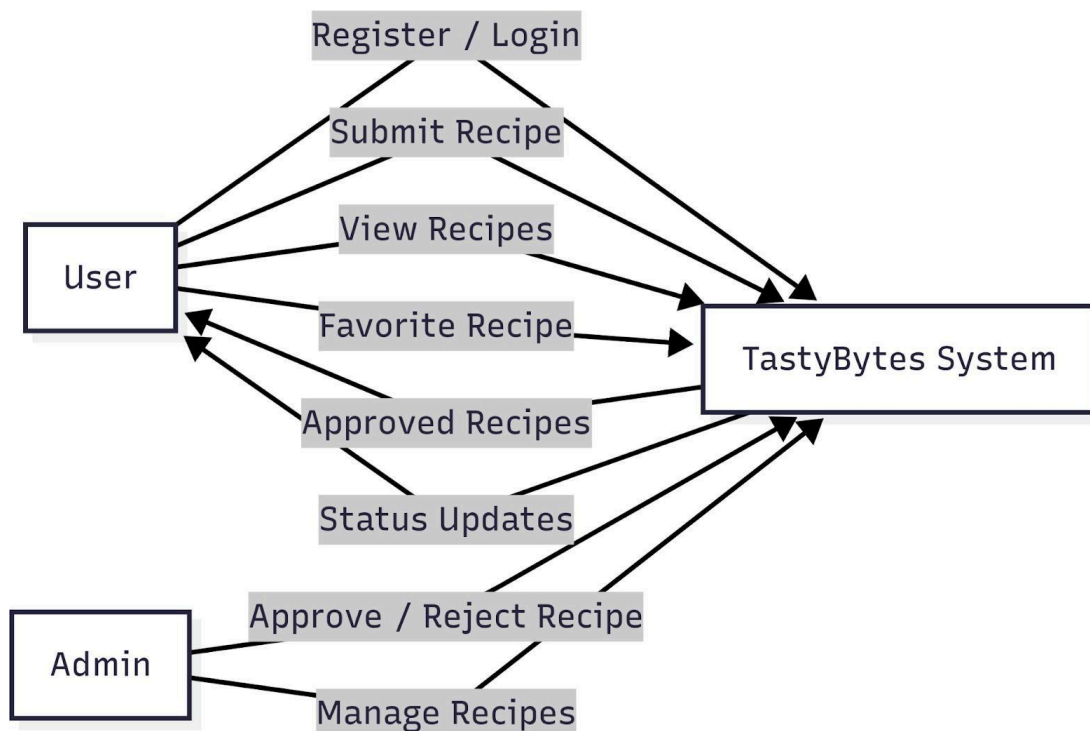
5.3 Use Cases Diagram



DATA FLOW DIAGRAMS (DFD)

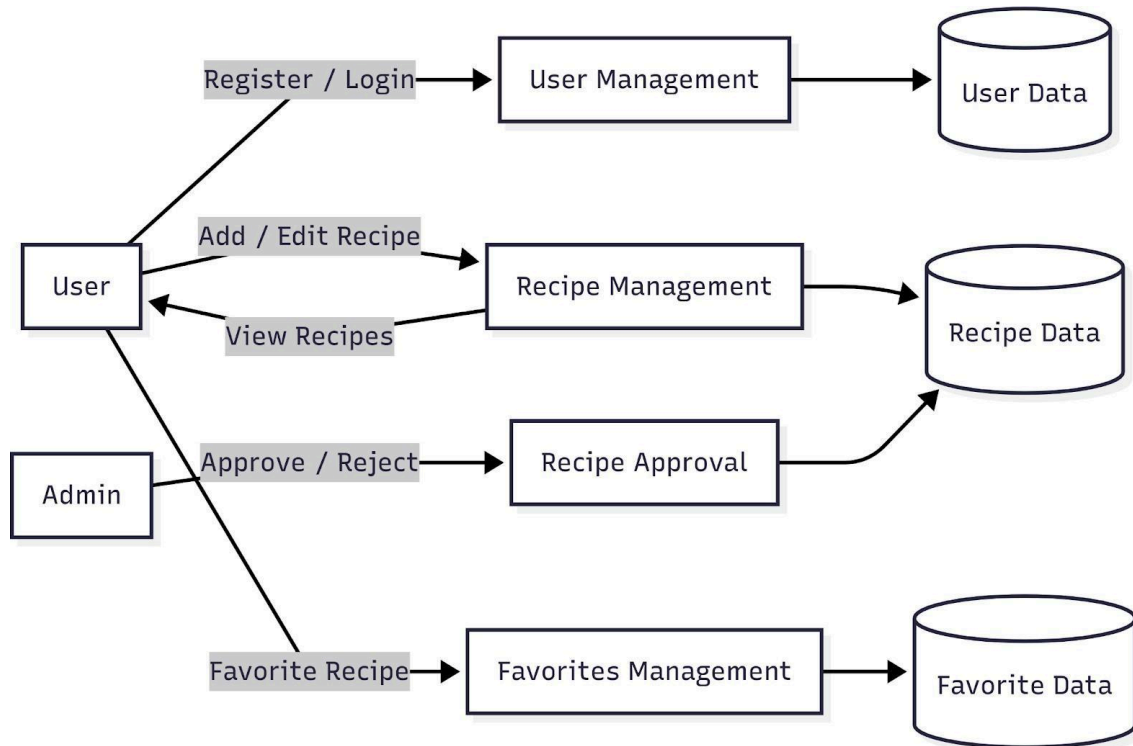
DFD Level 0 – Context Diagram Explanation

The Level 0 Data Flow Diagram represents the TastyBytes system as a single process and shows how it interacts with external entities. Users interact with the system to register, log in, submit recipes, view approved recipes, and manage favorites. Administrators interact with the system to approve or reject recipes and manage overall content. This diagram defines the system boundary and provides a high-level overview of data entering and leaving the system.



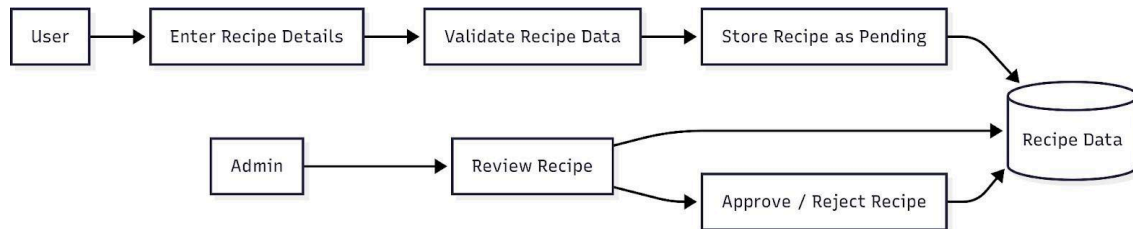
DFD Level 1 – Major Processes Explanation

The Level 1 Data Flow Diagram decomposes the TastyBytes system into its main functional processes. User Management handles registration and authentication. Recipe Management manages the creation, editing, viewing, and storage of recipes. Recipe Approval allows administrators to approve or reject submitted recipes. Favorites Management enables users to mark and view favorite recipes. Data is stored in separate data stores for users, recipes, and favorites, ensuring organized and structured data handling.



DFD Level 2 – Recipe Submission and Approval Explanation

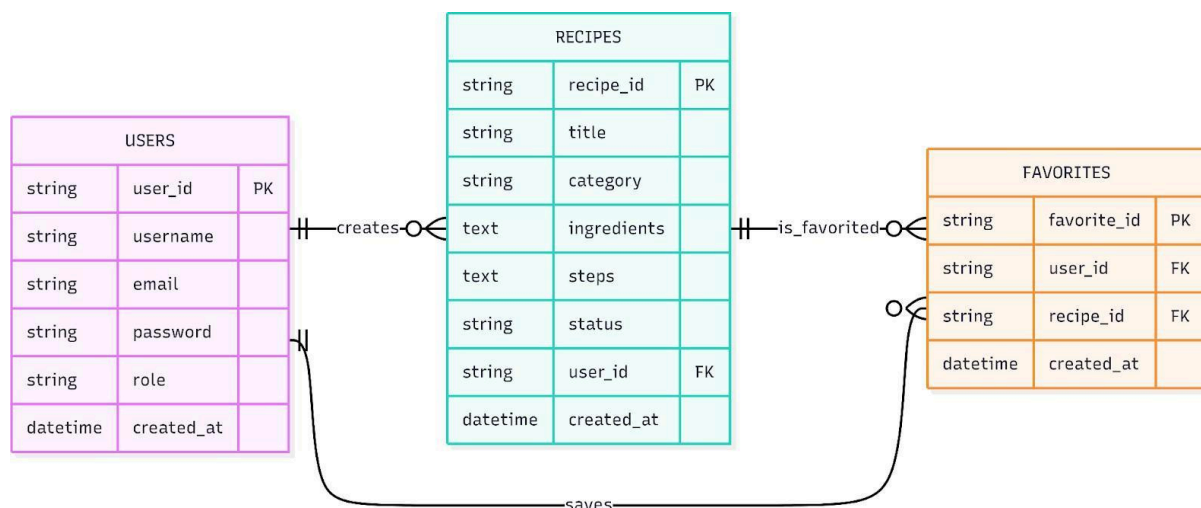
The Level 2 Data Flow Diagram provides a detailed view of the recipe submission and approval process. The user enters recipe details, which are validated by the system before being stored as a pending recipe. The administrator reviews the submitted recipe and decides whether to approve or reject it. The final decision updates the recipe status in the data store. This level illustrates the internal steps involved in managing recipe content quality.



PHASE 2: DESIGN PHASE

7. Database Design schema

The database schema for the TastyBytes Recipe Management System is designed to store and manage users, recipes, and favorite relationships in a structured and efficient manner. The schema follows basic normalization principles to reduce data redundancy and maintain data integrity.



7.1 Data Storage Structure (Simulated)

Entity: USER

Fields:

- user_id
- username
- email
- password
- role
- created_at

Entity: RECIPE

Fields:

- recipe_id
- title
- category
- ingredients
- steps
- image_url
- status
- author_id
- created_at

Entity: FAVORITE

Fields:

- favorite_id
- user_id
- recipe_id
- created_at

7.2 Relationships

- User creates many Recipes
- User favorites many Recipes
- Admin approves Recipes

7.3 Data Integrity

- Unique recipe titles enforced logically
- Role-based access validation

7.3 Database Statement

```
CREATE TABLE USERS (  
    user_id VARCHAR(50) PRIMARY KEY,  
    username VARCHAR(50) NOT NULL,  
    email VARCHAR(100) UNIQUE NOT  
    NULL, password VARCHAR(255) NOT  
    NULL,  
    role VARCHAR(20) NOT NULL,  
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

```
CREATE TABLE RECIPES (  
    recipe_id VARCHAR(50) PRIMARY KEY,  
    title VARCHAR(100) NOT NULL,  
    category VARCHAR(50) NOT NULL,  
    ingredients TEXT NOT NULL,  
    steps TEXT NOT NULL,  
    status VARCHAR(20) NOT NULL,  
    user_id VARCHAR(50) NOT NULL,  
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (user_id) REFERENCES USERS(user_id)  
);
```

```
CREATE TABLE FAVORITES (  
    favorite_id VARCHAR(50) PRIMARY KEY,  
    user_id VARCHAR(50) NOT NULL,
```

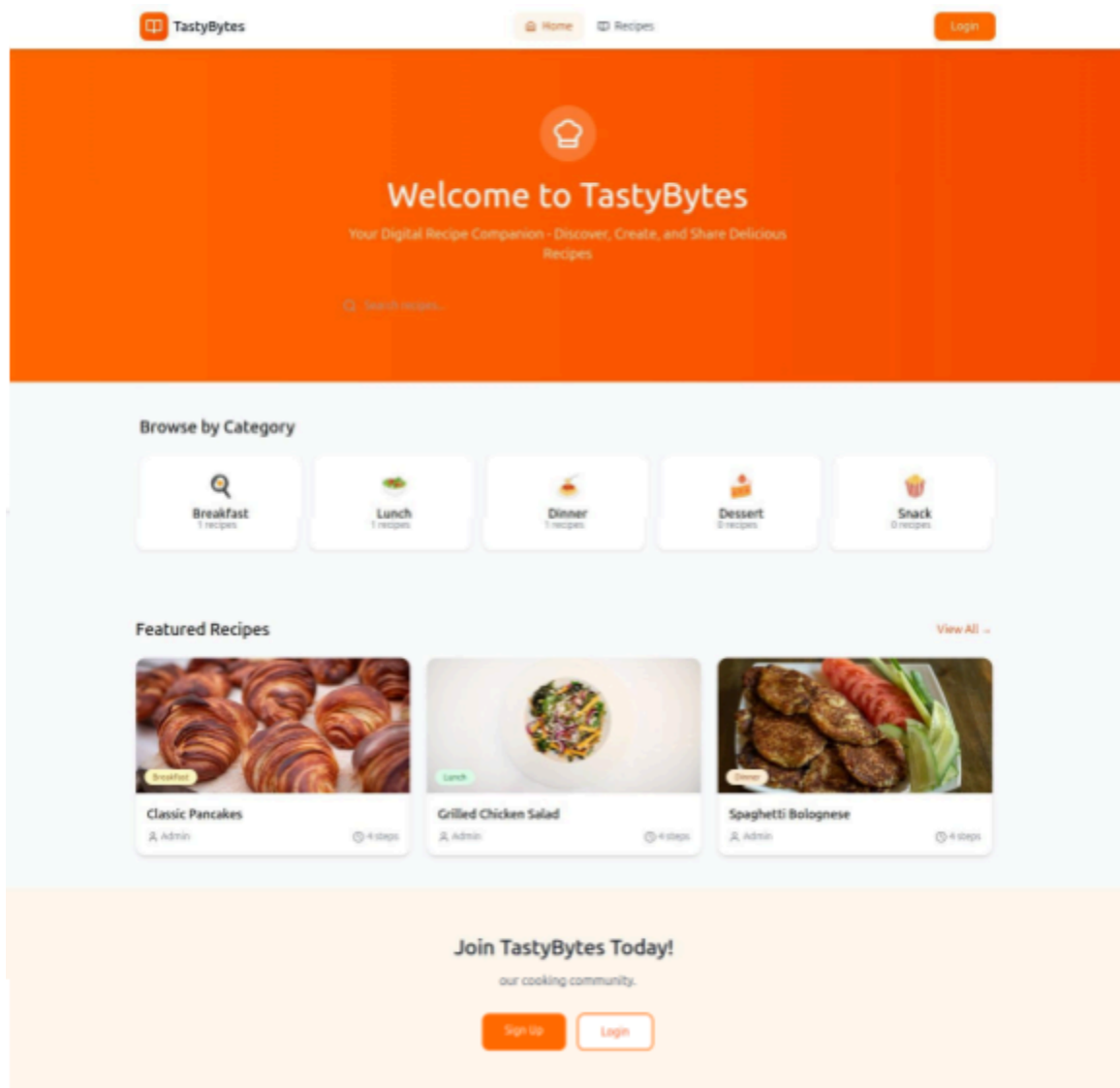
```
recipe_id VARCHAR(50) NOT NULL,  
created_at DATETIME DEFAULT CURRENT_TIMESTAMP,  
FOREIGN KEY (user_id) REFERENCES USERS(user_id),  
FOREIGN KEY (recipe_id) REFERENCES RECIPES(recipe_id),  
UNIQUE (user_id, recipe_id)  
);
```

8. User Interface Overview

The interface is clean, responsive, and optimized for recipe browsing and management.

Key Pages:

- Home Page



- Login Page

 TastyBytes

[Home](#) [Recipes](#)

Login



Welcome Back!

Log in to your TastyBytes account

Email Address

 you@example.com

Password

 *****

Log In

Don't have an account? [Sign up](#)

Demo accounts:
[Admin: admin@tastybytes.com / admin123](#)

- Register Page

 TastyBytes

[Home](#) [Recipes](#)

Login



Join TastyBytes

Create your account and start cooking

Username

 johndoe

Email Address

 you@example.com

Password

 *****

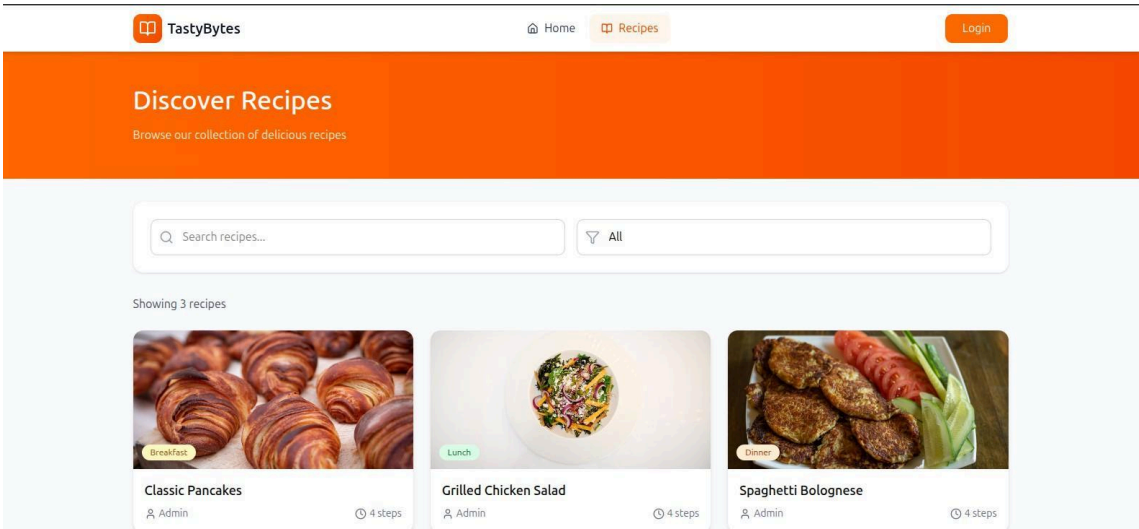
Confirm Password

 *****

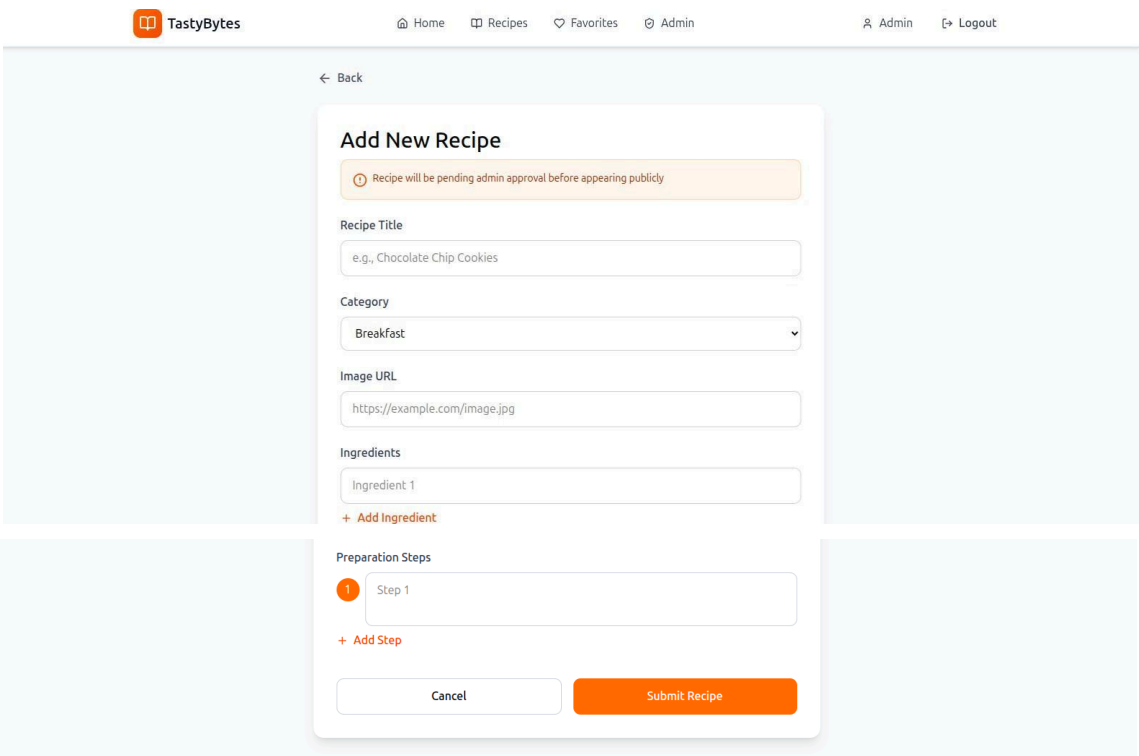
Create Account

Already have an account? [Log in](#)

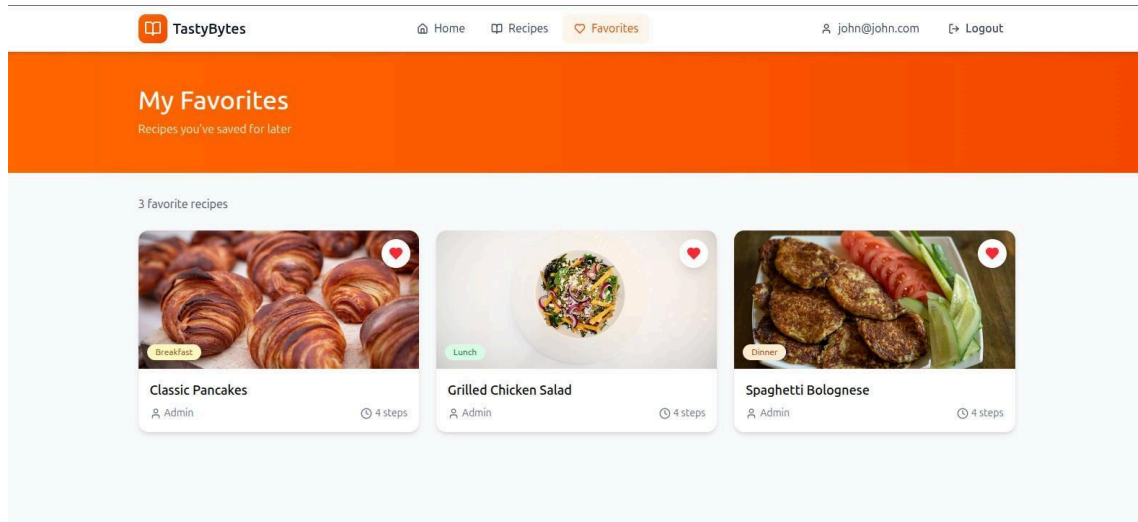
• Recipe Dashboard



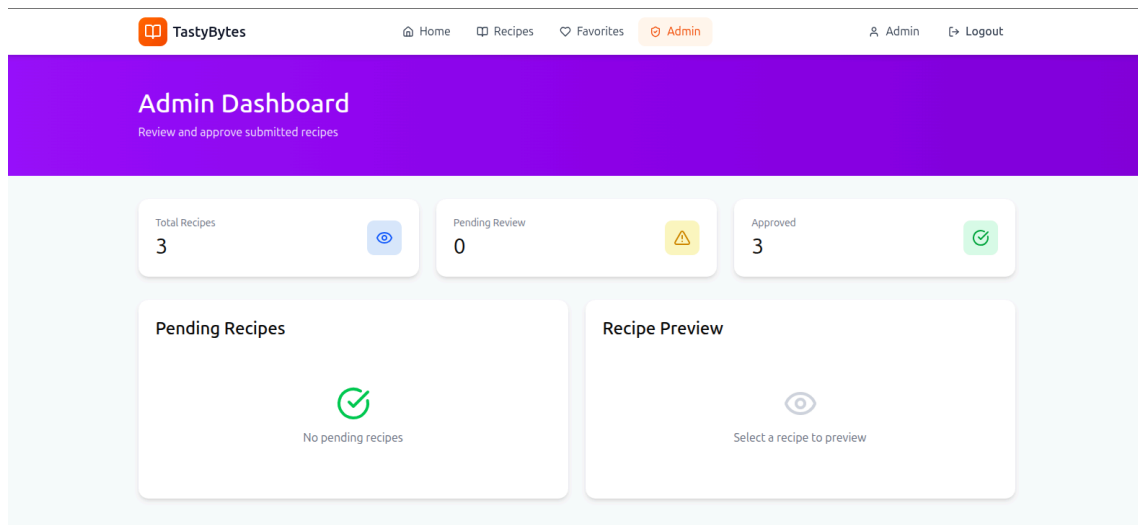
• Add/Edit Recipe Page



- Favorites Page



- Admin Dashboard

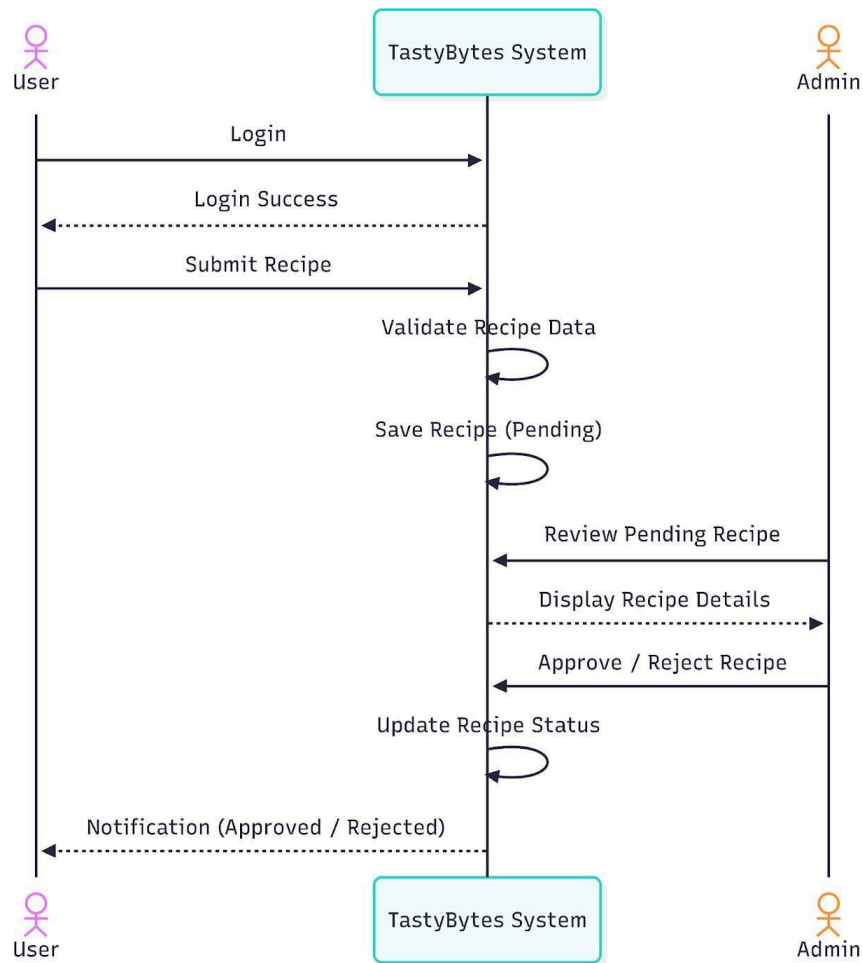


9. Sequence Diagrams

The sequence diagram illustrates the interaction between the user, the TastyBytes system, and the administrator during the recipe submission and approval process. First, the user logs into the system and submits a new recipe. The system validates the recipe details and stores it with a pending status. The administrator then reviews the submitted recipe and decides to approve or reject it. Finally, the system updates the recipe status and notifies the user of the decision. This diagram shows the order of interactions and responsibilities among system actors.

Key Flows:

- User Registration
- Login Authentication
- Recipe Submission
- Admin Approval Process
- Favorite Recipe Flow



10. UML Diagrams

Unified Modeling Language (UML) diagrams are used to visually represent the structure and behavior of the TastyBytes Recipe Management System. UML helps developers and stakeholders understand how system components interact and how responsibilities are distributed among users, administrators, and system entities.

